



PROJECT MAIEUTIC / 學問

From Inquiry to Software

The CityUHK Initiative on
Specification-Driven & Agentic Computing Education

“The art of bringing
latent knowledge
into light through
questioning.”



QUESTION
Inquire deeply



SPECIFY
Constrain clearly



TEACH THE AGENT
Dialogue & guide



BUILD
Agents create



REFLECT
Learn & improve



FROM FIRST YEAR ONWARD,
STUDENTS LEARN TO
**THINK, SPECIFY, AND
BUILD WITH AI AGENTS.**

★ FIRST-YEAR PROJECT

SHIPPED TO PRODUCTION

slideshow.

Turning a Jupyter notebook into a live, animated slideshow —
with no export step.

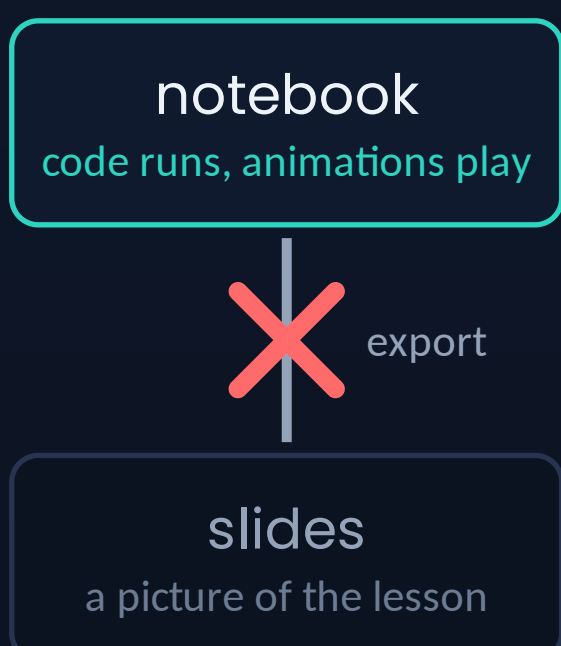
A JupyterLab extension that presents a notebook's own cells as a Reveal.js slideshow. The cells stay attached to the kernel, so code still runs mid-lecture, SVG animations play in both views, and an equation can be revealed a piece at a time. Nothing is exported: edit a cell and the slide has already changed.

DOMAIN	Computing education · presentation tooling
TECH STACK	TypeScript, JupyterLab extension API, Reveal.js, svg.js, MathJax 4
METHOD	Specification first — each feature written as a spec, then tested in a live lecture
SHIPPED	PyPI · MIT licence · deployed on DIVE for CS1302
OUTCOME	One of the student packages behind “DIVE into Agentic Coding”, accepted at IEEE TALE 2026 [1]

01 THE QUESTION

Why must a lecturer keep two copies of the same lesson?

Exporting freezes the lesson: code stops running, animations become GIFs, and the deck drifts from the notebook students are given.



Not “how do I export better?” but “**why export at all?**” — the question that became the specification.

02 THE ARTEFACT

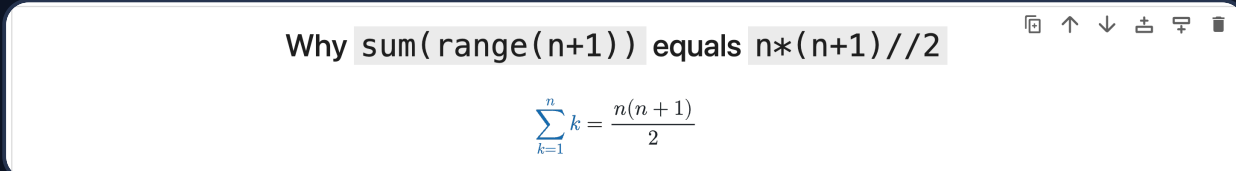
ONE MARKDOWN CELL — THE SOURCE

```
<div data-animate>
<h2>Why the sum equals  $n(n+1)/2$ </h2>

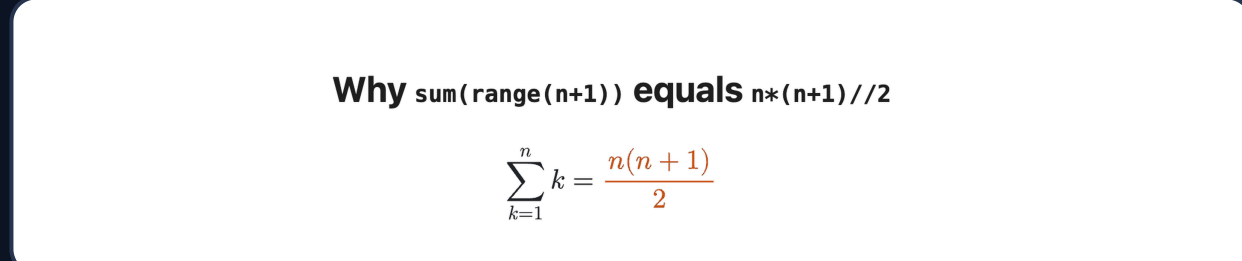
$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

<!-- { "animation": {
  { "element": ".lhs", "modifier": "attr",
    "parameters": [{ "fill": "#1c7ed6" }], "duration": 600 },
  { "element": ".rhs", "modifier": "attr",
    "parameters": [{ "fill": "#e8590c" }], "duration": 600 },
  }, "loop": true } -->
</div>
```

IN THE NOTEBOOK



AS A SLIDE



One cell, animating in both views. Vector SVG, not a GIF — and MathJax 4 lets one side of an equation be highlighted on its own.

03 HOW IT WORKS

- Read the raw cell**
JupyterLab strips animated SVG; the extension reads the source itself
- Move, never copy**
Live cell nodes move into Reveal.js sections; the kernel survives
- Drive the animation**
One timeline, one config, both views
- Hand back the keys**
Shift+Enter mid-lecture: output lands on the slide
The slide is the cell.
Nothing is exported, so nothing can go stale.

04 THE MAIEUTIC LOOP, IN PRACTICE

Every feature began as a specification, not as code. One real cycle: a package that installed cleanly and then did nothing.

QUESTION

Installs cleanly, then does nothing. Why?

SPECIFY

One plugin id, one schema path, one build.

TEACH THE AGENT

Give it the console error and both file trees; ask what differs.
`Token ... separate declarations`

REFLECT

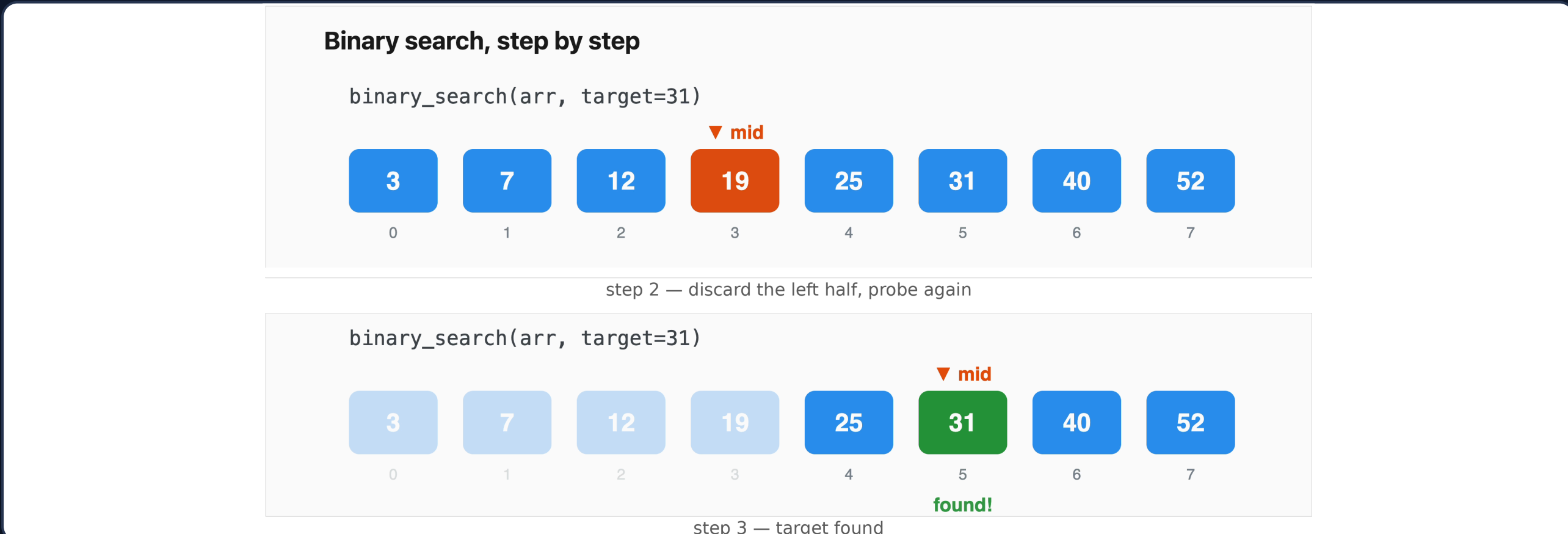
Two copies of one dependency. The lesson outlived the bug.

the answer becomes the next question

Twelve releases came out of that loop, each one starting with a question from a real lecture.

06 WHAT AN ANIMATION LOOKS LIKE

A few lines of configuration, and an algorithm explains itself — on the page and on the projector.



07 TRY IT NOW

No install, no account. Each code opens that notebook and starts it.



Feature tour
Scrollable slides, incremental reveal, live code



Animated maths
One side of an equation highlighted at a time



A real lecture
92 cells of CS1302, presented as it is taught

`pip install slideshow`



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REFERENCES

- [1] C. Zhao, Z. Wang, **A. A. Ahmedani**, B. Karimtayev, Marcello-Raphael-Christiano, N. M. B. Lua, R. Khamza, and C. Chan, “DIVE into Agentic Coding: Can You Teach AI to Teach You?” in *Proc. 2026 IEEE Int. Conf. on Teaching, Assessment and Learning for Engineering (TALE)*, 2026, accepted.
- [2] C. Chan et al., “DIVE: Make Online Learning Diversified, Interactive, Versatile, and Engaging,” in *2022 IEEE TALE*, 2022, pp. 482–489, doi: 10.1109/TALE54877.2022.00085.
- [3] H. El Hattab and contributors, “Reveal.js — the HTML presentation framework.” revealjs.com.
- [4] A. Goel, “reveal.js-plugins: animate,” github.com/rajgoel/reveal.js-plugins.
- [5] Project Jupyter, “JupyterLab documentation,” jupyterlab.readthedocs.io.



Source code



LinkedIn



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